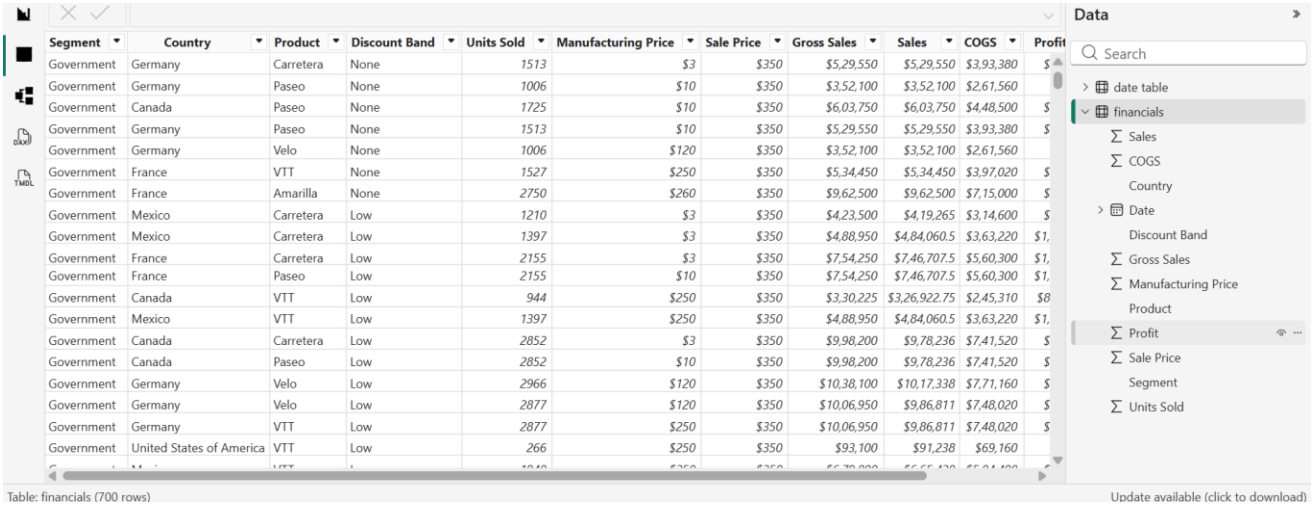
Data Analytics (DA)

“Data Transformation, Modeling & Visualization with Power BI”

TITLE: EXECUTIVE FINANCIAL ANALYSIS USING POWER BI

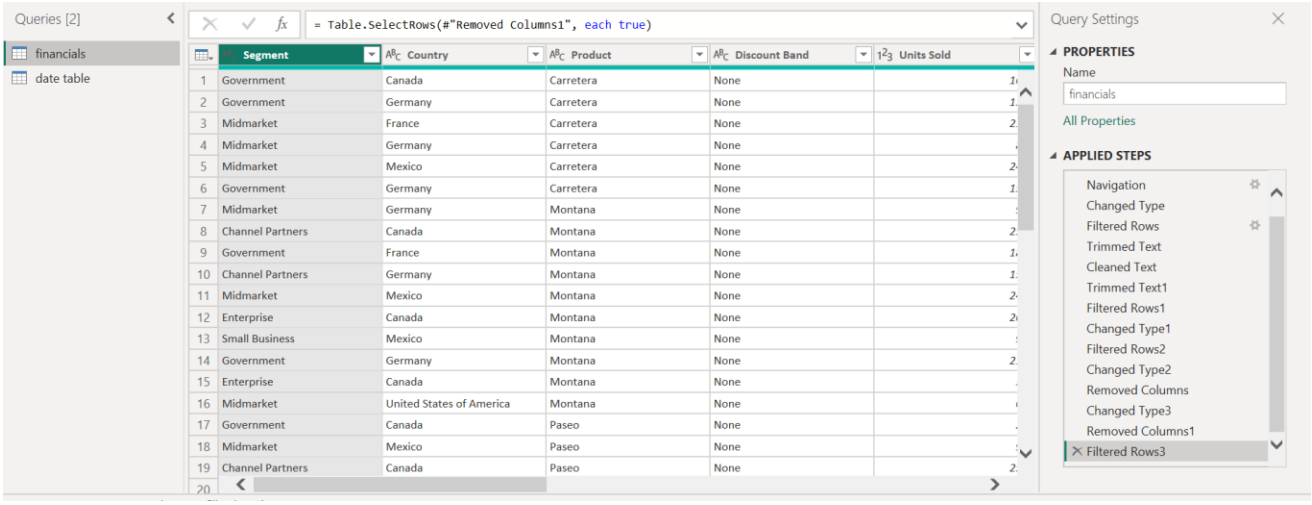
DATA CONNECTION, IMPORT, INTERFACE EXPLORATION



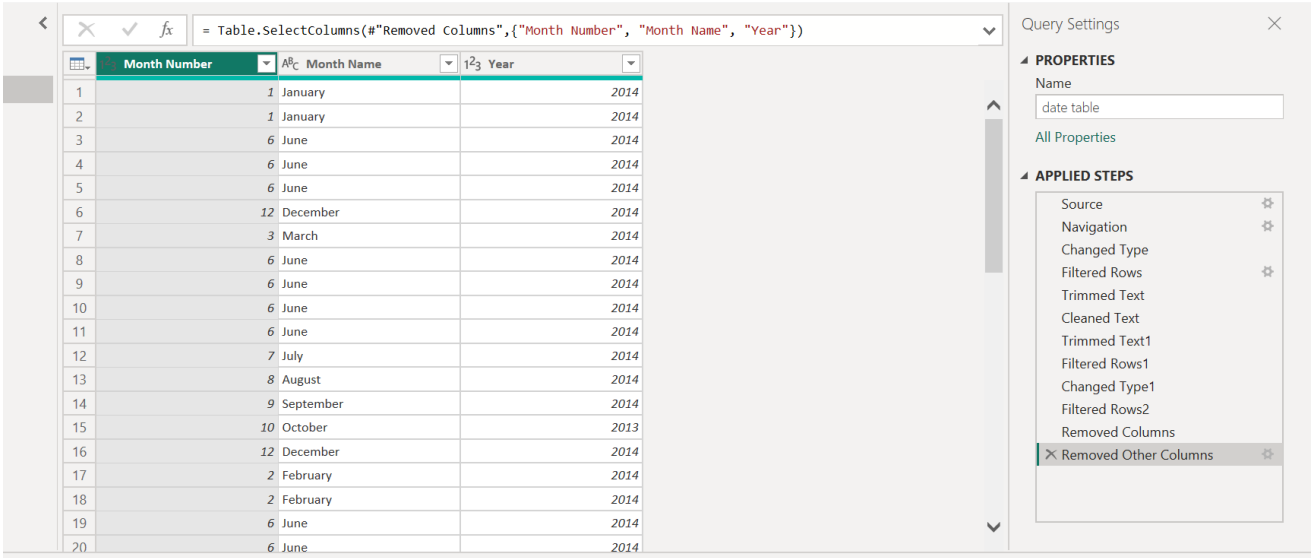
Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Sales	COGS	Profit
Government	Germany	Carretera	None	1513	$3	$350	$5,29,550	$5,29,550	$3,93,380	$
Government	Germany	Paseo	None	1006	$10	$350	$3,52,100	$3,52,100	$2,61,560	
Government	Canada	Paseo	None	1725	$10	$350	$6,03,750	$6,03,750	$4,48,500	$
Government	Germany	Paseo	None	1513	$10	$350	$5,29,550	$5,29,550	$3,93,380	$
Government	Germany	Velo	None	1006	$120	$350	$3,52,100	$3,52,100	$2,61,560	
Government	France	VTT	None	1527	$250	$350	$5,34,450	$5,34,450	$3,97,020	$
Government	France	Amarilla	None	2750	$260	$350	$9,62,500	$9,62,500	$7,15,000	$
Government	Mexico	Carretera	Low	1210	$3	$350	$4,23,500	$4,19,265	$3,14,600	$
Government	Mexico	Carretera	Low	1397	$3	$350	$4,88,950	$4,84,060.5	$3,63,220	$1,
Government	France	Carretera	Low	2155	$3	$350	$7,54,250	$7,46,707.5	$5,60,300	$1,
Government	France	Paseo	Low	2155	$10	$350	$7,54,250	$7,46,707.5	$5,60,300	$1,
Government	Canada	VTT	Low	944	$250	$350	$3,30,225	$3,26,922.75	$2,45,310	$8
Government	Mexico	VTT	Low	1397	$250	$350	$4,88,950	$4,84,060.5	$3,63,220	$1,
Government	Canada	Carretera	Low	2852	$3	$350	$9,98,200	$9,78,236	$7,41,520	$
Government	Canada	Paseo	Low	2852	$10	$350	$9,98,200	$9,78,236	$7,41,520	$
Government	Germany	Velo	Low	2966	$120	$350	$10,38,100	$10,17,338	$7,71,160	$
Government	Germany	Velo	Low	2877	$120	$350	$10,06,950	$9,86,811	$7,48,020	$
Government	Germany	VTT	Low	2877	$250	$350	$10,06,950	$9,86,811	$7,48,020	$
Government	United States of America	VTT	Low	266	$250	$350	$93,100	$91,238	$69,160	

DATA CLEANING & TRANSFORMATION

- Formatted & Transformed Columns in Power Query

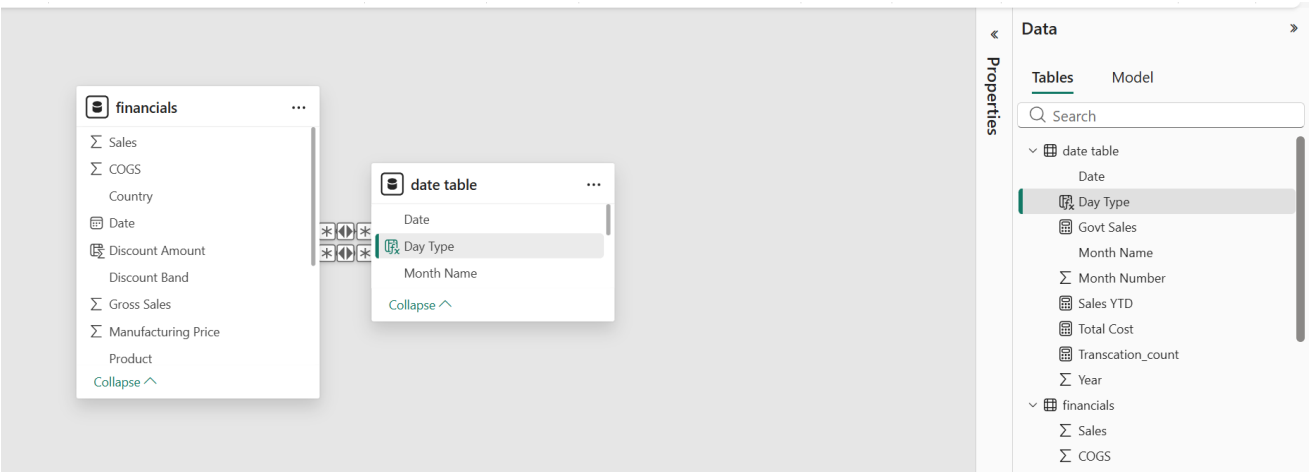


- Date table created



Data Modeling

- Built Tables, Defined Relationships



DAX & CALCULATIONS

✓ **Created calculated measures:**

✓ `Total Units = SUM(financials[Units Sold])`

✓ `Govt Sales = CALCULATE(SUM(financials[ Sales]), financials[Segment] = "Government")`

✓ `Total Cost = SUM(financials[COGS])`

✓ `Sales YTD = TOTALYTD(SUM(financials[ Sales]), 'date table'[Date])`

✓ `Transcation_count = DISTINCTCOUNT(financials[Product])`

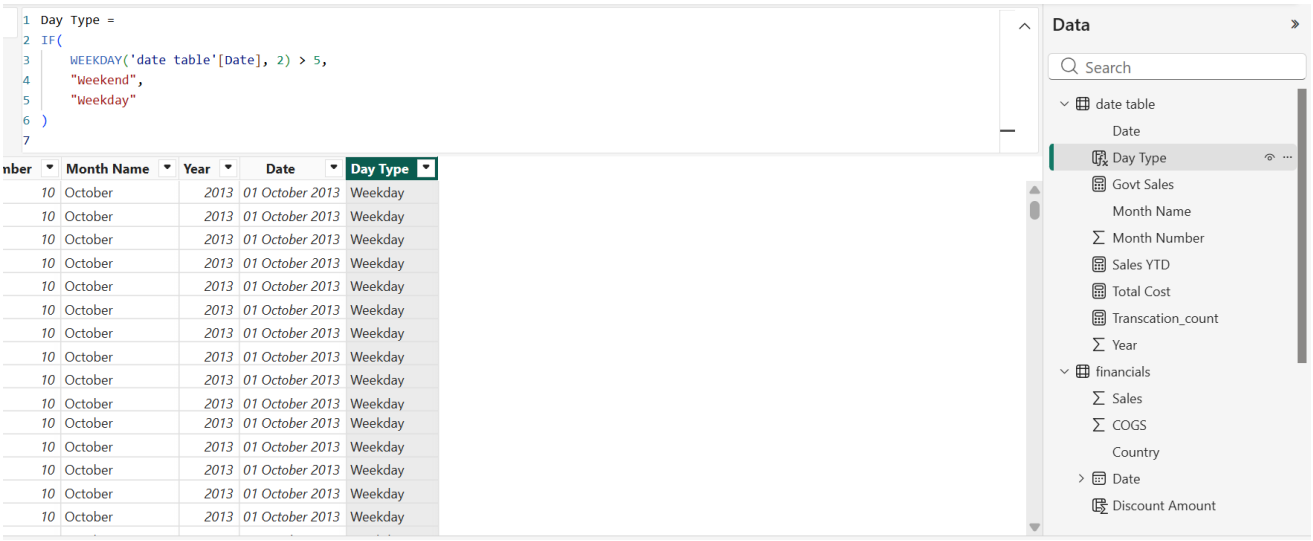
Total Units	Transcation_count	Total Cost	Sales YTD	Govt Sales
1,126K	6	₹ 102M	₹ 92M	₹ 52,504K

Descriptive Statistics

- **Total Units (1,126K):** This represents the **Aggregate Sum ()** of the units sold. In descriptive terms, it defines the **Total Volume** of physical inventory moved during the specified period, establishing the scale of business operations.
- **Transaction count (6):** This is a **Frequency Count of Unique Categories**. Statistically, this represents the count of distinct time periods (such as months or years) present in the data, indicating the **Temporal Breadth** of the dataset being analyzed.
- **Total Cost (₹102M):** This represents the **Cumulative Sum** of all expenses (COGS). Descriptively, it quantifies the **Total Resource Outflow** required to generate the current business volume, serving as the primary baseline for expenditure analysis.
- **Sales YTD (₹92M):** This is a **Time-Series Aggregate**. It measures the **Cumulative Frequency of Revenue** from the start of the fiscal year to the current date, providing a benchmark for year-to-date performance against annual targets.
- **Govt Sales (₹52,504K):** This is a **Filtered Sub-total**. It describes the **Segmented Contribution** of the Government sector to the total revenue. Statistically, it shows that this single segment drives more than 50% of the Year-to-Date sales.

CREATED CALCULATED COLUMNS

"DAY TYPE" column created



“**DISCOUNT AMOUNT**” column created

```
1 Discount Amount = 'financials'[Sale Price] * 0.10
```

s Sold	Manufacturing Price	Sale Price	Gross Sales	Sales	COGS	Profit	Date	Discount Amount	Transaction Type
1513	$3	$350	$5,29,550	$5,29,550	$3,93,380	$1,36,170	01-12-2014	₹ 35	High Value
1006	$10	$350	$3,52,100	$3,52,100	$2,61,560	$90,540	01-06-2014	₹ 35	High Value
1725	$10	$350	$6,03,750	$6,03,750	$4,48,500	$1,55,250	01-11-2013	₹ 35	High Value
1513	$10	$350	$5,29,550	$5,29,550	$3,93,380	$1,36,170	01-12-2014	₹ 35	High Value
1006	$120	$350	$3,52,100	$3,52,100	$2,61,560	$90,540	01-06-2014	₹ 35	High Value
1527	$250	$350	$5,34,450	$5,34,450	$3,97,020	$1,37,430	01-09-2013	₹ 35	High Value
2750	$260	$350	$9,62,500	$9,62,500	$7,15,000	$2,47,500	01-02-2014	₹ 35	High Value
1210	$3	$350	$4,23,500	$4,19,265	$3,14,600	$1,04,665	01-03-2014	₹ 35	High Value
1397	$3	$350	$4,88,950	$4,84,060.5	$3,63,220	$1,20,840.5	01-10-2014	₹ 35	High Value
2155	$3	$350	$7,54,250	$7,46,707.5	$5,60,300	$1,86,407.5	01-12-2014	₹ 35	High Value
2155	$10	$350	$7,54,250	$7,46,707.5	$5,60,300	$1,86,407.5	01-12-2014	₹ 35	High Value
944	$250	$350	$3,30,225	$3,26,922.75	$2,45,310	$81,612.75	01-04-2014	₹ 35	High Value
1397	$250	$350	$4,88,950	$4,84,060.5	$3,63,220	$1,20,840.5	01-10-2014	₹ 35	High Value
2852	$3	$350	$9,98,200	$9,78,236	$7,41,520	$2,36,716	01-12-2014	₹ 35	High Value
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2966	$120	$350	$10,38,100	$10,17,338	$7,71,160	$2,46,178	01-10-2013	₹ 35	High Value
2877	$120	$350	$10,06,950	$9,86,811	$7,48,020	$2,38,791	01-10-2014	₹ 35	High Value
2877	$250	$350	$10,06,950	$9,86,811	$7,48,020	$2,38,791	01-10-2014	₹ 35	High Value
266	$250	$350	$93,100	$91,238	$69,160	$22,078	01-12-2013	₹ 35	High Value

Data

Search

- Month Name
- Month Number
- Sales YTD
- Total Cost
- Transcation_count
- Year
- financials
  - Sales
  - COGS
  - Country
  - Date
  - Discount Amount
  - Discount Band
  - Gross Sales
  - Manufacturing Price
  - Product

)0 rows) Column: Discount Amount (7 distinct values)

Update available (click to download)

“**TRANSACTION TYPE** “column created

```
1 Transaction Type =
2 IF(
3     financials[Gross Sales] > 10000,
4     "High Value",
5     "Standard"
6 )
7
```

old	Manufacturing Price	Sale Price	Gross Sales	Sales	COGS	Profit	Date	Discount Amount	Transaction Type
1513	$3	$350	$5,29,550	$5,29,550	$3,93,380	$1,36,170	01-12-2014	₹ 35	High Value
1006	$10	$350	$3,52,100	$3,52,100	$2,61,560	$90,540	01-06-2014	₹ 35	High Value
1725	$10	$350	$6,03,750	$6,03,750	$4,48,500	$1,55,250	01-11-2013	₹ 35	High Value
1513	$10	$350	$5,29,550	$5,29,550	$3,93,380	$1,36,170	01-12-2014	₹ 35	High Value
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2852	$3	$350	$9,98,200	$9,78,236	$7,41,520	$2,36,716	01-12-2014		

Name 'financials'[Transaction Type]

Data

Search

- financials
  - Sales
  - COGS
  - Country
  - Date
  - Discount Amount
  - Discount Band
  - Gross Sales
  - Manufacturing Price
  - Product
  - Profit
  - Sale Price
  - Segment
  - Total Units
  - Transaction Type
  - Units Sold

ows) Column: Transaction Type (2 distinct values)

Update available (click to download)

DATA VISUALIZATION

INSIGHTS:

1.Key Performance Indicators (KPIs)

- Total Volume: The business moved 1,126K units, indicating high market penetration and operational scale.
- Revenue vs. Expenditure: Total Sales YTD stands at ₹92M, while Total Cost is ₹102M. This reveals a ₹10M deficit, suggesting that current production or operational costs are outpacing revenue.
- Core Revenue Driver: Government Sales (₹52,504K) account for approximately 57% of total Year-to-Date sales, identifying the public sector as the primary revenue stream.

2. Profitability by Segment

- Top Performer: The Government segment is the most profitable, contributing $11.39M.
- Critical Underperformer: The Enterprise segment is currently operating at a loss of $0.61M. This is a priority area for cost restructuring or pricing adjustments.
- Supporting Segments: Small Business ($4.14M) and Channel Partners ($1.32M) remain healthy contributors to the bottom line.

3. Operational & Sales Trends

- Temporal Growth: The Sales by Year donut chart shows that 2014 saw a massive expansion compared to 2013, representing the majority of the current dataset's value.
- Daily Activity: Weekdays consistently outperform weekends in terms of units sold.
- Discount Strategy: High-volume sales are largely driven by "None" or "Low" discount bands. This suggests that the product maintains strong demand without requiring aggressive price slashing.

4. Geographic Cost Distribution

- The global map indicates that **Total Cost** is heavily concentrated in **North America and Europe**. While these are major markets, the high cost-to-revenue ratio in these regions may be contributing to the overall ₹10M deficit.

Conclusion

- ✓ Cost Optimization: Investigate the ₹10M gap between Cost and Sales, focusing on high-cost regions identified in the map.
- ✓ Enterprise Pivot: Analyse the Enterprise segment loss to determine if the issue lies in high manufacturing prices or insufficient sale prices.
- ✓ Capitalize on Momentum: Leverage the high-volume, low-discount success seen on Weekdays to further reduce reliance on price-cutting strategies.